

■ EXCEL FOR MACHINE LEARNING (ML) – MASTER NOTES

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1. DATA CLEANING ★★★★★

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Purpose:

Data ko clean aur consistent banana.

Functions:

- TRIM() → Extra spaces remove
- CLEAN() → Hidden characters remove
- UPPER() → Uppercase
- LOWER() → Lowercase
- PROPER() → First letter capital
- TEXT TO COLUMNS
- Remove Duplicates
- Find & Replace

Practice:

- ✓ Remove extra spaces
- ✓ Remove duplicates
- ✓ Convert names to Proper Case
- ✓ Split Full Name into First & Last Name

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2. MISSING VALUES ★★★★★

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Purpose:

Blank values ko identify aur handle karna.

Functions:

- COUNTBLANK()
- ISBLANK()

Methods:

- Fill with Mean
- Fill with Median
- Fill with Mode
- Delete rows (if very few blanks)

Practice:

- ✓ Count blank cells
- ✓ Fill missing values
- ✓ Compare Mean vs Median filling

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3. BASIC STATISTICS ★★★★★

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Functions:

- AVERAGE()
- MEDIAN()
- MODE.SNGL()
- MIN()
- MAX()
- COUNT()

- COUNTA()
- STDEV.S()
- VAR.S()

Remember:

Mean = Average

Median = Middle Value

Mode = Most Frequent Value

Standard Deviation = Data Spread

Variance = Spread²

Practice:

- ✓ Find Mean
- ✓ Find Median
- ✓ Find Mode
- ✓ Find Standard Deviation
- ✓ Find Variance

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4. SORTING & FILTERING ★★★★★

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Topics:

- Ascending Sort
- Descending Sort
- Custom Sort
- Filter
- Multiple Filter

Practice:

- ✓ Highest Salary
- ✓ Lowest Salary
- ✓ Employees from Mumbai
- ✓ Age > 30

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5. CONDITIONAL FUNCTIONS ★★★★★

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Functions:

- IF()
- IFS()
- AND()
- OR()
- IFERROR()

Practice:

- ✓ Pass / Fail
- ✓ High / Low Salary
- ✓ Eligible / Not Eligible

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6. LOOKUP FUNCTIONS ★★★★★

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Functions:

- XLOOKUP()
- VLOOKUP()

- INDEX()
- MATCH()

Practice:

- ✓ Employee ID → Salary
- ✓ Product ID → Price
- ✓ Student Roll No → Marks

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7. PIVOT TABLE ★★★★★

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Practice:

- ✓ Region Wise Sales
- ✓ Employee Wise Sales
- ✓ Product Wise Sales
- ✓ Monthly Sales

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8. CHARTS ★★★★★

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Charts:

- Bar Chart
- Line Chart
- Pie Chart
- Histogram
- Scatter Plot ★★★★★

Practice:

- ✓ Sales vs Profit
- ✓ Age Distribution
- ✓ Monthly Sales Trend

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9. CORRELATION ★★★★★

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Purpose:

Relationship between two variables.

Examples:

- ✓ Height vs Weight
- ✓ Experience vs Salary
- ✓ Study Hours vs Marks

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10. NORMALIZATION ★★★★★

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Methods:

- Min-Max Scaling
- Z-Score Standardization

Purpose:

Bring data to the same scale before ML.

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TOP EXCEL FUNCTIONS FOR ML

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TRIM()

CLEAN()

PROPER()

UPPER()

LOWER()

IF()

IFS()

AND()

OR()

IFERROR()

AVERAGE()

MEDIAN()

MODE.SNGL()

MIN()

MAX()

COUNT()

COUNTA()

COUNTBLANK()

ISBLANK()

STDEV.S()

VAR.S()

XLOOKUP()

VLOOKUP()

INDEX()

MATCH()

SUM()

SUMIF()
SUMIFS()
COUNTIF()
COUNTIFS()

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DAILY 30–40 MIN PRACTICE

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1. Data Cleaning
2. Missing Values
3. Statistics
4. IF Functions
5. Lookup
6. Pivot Table
7. Charts
8. Correlation

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INTERVIEW REMEMBER

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- ✓ Data Cleaning
- ✓ Missing Values
- ✓ Mean, Median, Mode
- ✓ Standard Deviation
- ✓ Variance
- ✓ Sorting & Filtering
- ✓ Pivot Table

- ✓ Lookup Functions
- ✓ Correlation
- ✓ Normalization
- ✓ Scatter Plot
- ✓ Histogram